The Macroeconomic Effects of Lifting Barriers to Migration

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NEUDC

Stark Urban–Rural Gap in Low-Income Countries





Pictures show a rural Kenyan home and Nairobi, the capital city of Kenya. Source: Jocelyn Diaz (left) and Catalin Marin (right).

Returns to Migration Are High

- huge urban–rural gaps in income
  - Kenya: **3.3x urban–rural productivity gap** (URPG)
  - large **marginal income returns** to migration (Bryan, Chowdhury, Mobarak, 2014)
- urbanization remains limited
- growing randomized controlled trial evidence on why!

Why Do Gaps Persist and What Can We Do? RCT Evidence

- migration costs, fin. constraints *(Bryan, Chowdhury, Mobarak, 2014)*
  - provide cash to cover migration cost
- unemployment risk *(Miner, 2026)*
  - provide urban unemployment insurance
- underestimation of urban income *(Baseler, 2023; Barnett-Howell et al., 2026)*
  - provide information about urban earnings

which barriers are costliest?
which interventions are most effective?

- unclear because experiments have
  - different environment and sampling
  - different intervention design and measurement
  - models tailored to specific experiment

This Paper

- build and estimate a quantitative general equilibrium model of internal migration to match moments from Kenyan data
- reproduce several prominent experiments in Kenya within a single environment
  - approximate sampling, intervention, and measurement of each without targeting treatment effects
- model matches some migration treatment effects, misses others
- long-run effects can differ substantially from experimental endlines
- national rollout of UI, information interventions modestly effective
- decompose the urban–rural income gap into contributions from different frictions:
  - information is the strongest explicit friction: explains **19%** of the URPG
  - residual migration costs explain **87%** of the URPG

Model

Model Overview

Aiyagari-style dynamic heterogeneous agent model with migration

Regions

- Rural (R) and Urban (U)

Heterogeneous Households Choose

- consumption c
- savings a'
- location l'

Households Heterogeneous In

- rural, urban productivity z_R, z_U ▶ *details*
- location l
- urban experience x ▶ *preferences*
- employment status e
- assets a

Frictions

- borrowing constraint: $a \geq \underline{a}$
- information friction:
  - rural households perceive $\tilde{w}_U = \gamma w_U$, $\gamma \in (0, 1]$
- urban unemployment risk:
  - job separation λ_s
  - job finding λ_f
- migration costs: pay m_c when moving
- migration disutility:
  - the inexperienced discount urban consumption by m_u

Reproducing Experiments

Reproducing Experiments: Strategy

- estimate the model to match moments from observational data
  - info friction, unemployment risk, urban experience map directly to observed moments
  - migration cost estimated to match low urban share
    - capturing all other migration obstacles

  ▶ *estimation* ▶ *moments*
- for each RCT intervention, create a model representation of:
  - sampling
  - treatment
    - matching some metric of treatment intensity, *not* treatment effects
    - leaves treatment effects to serve as non-targeted moments
  - measurement

Cash Transfer: Egger et al. (2022)

Sampling

RCT

- 1 rural Kenyan county
- half of villages treated
- eligibility: thatched roof (1/3 of households)

Model

- eligibility: poorest 1/3 of rural households by assets
- randomly treat half

Treatment

RCT

- one-time unconditional transfer
- 75% of mean annual expenditure

Model

- one-time asset transfer
- 75% of mean rural annual consumption

Measurement

RCT

- endline: 19 months after transfer
- migration and household expenditure, treatment vs control

Model

- endline: 3 periods (18 months)
- migration: urban residence at endline
- consumption

Cash Transfer: Egger et al. (2022) Results

	Migration Rate Change		Consumption Change	
Treatment arm	RCT	Model	RCT	Model
Cash transfer	1.0 pp (1.0)	0.0 pp	11.6%*** (2.4)	14.3%

- model matches negligible short-run migration response, magnitude of consumption response

▶ *alternative experiments*

Unemployment Insurance and Cash Transfer: Miner (2026)

Sampling

RCT

- male rural workers aged 18–30 in 1 rural county
- most had prior migration experience
- sampled 12 workers per village

Model

- sample rural households
- prioritize experienced rural households

Treatment

RCT

- UI: up to 15 days at average urban wage (60 USD)
- cash: one-time 50 USD transfer

Model

- UI: b matching 15 days of avg urban wage
- cash: asset transfer, 50/60 of UI amount

Measurement

RCT

- endline: about 6 months after treatment
- migration and income, treatment vs control

Model

- endline: 1 period (6 months)
- urban residence and income changes relative to rural baseline income

Unemployment Insurance and Cash Transfer: Miner (2026) Results

	Migration Rate Change		Income Change	
Treatment arm	RCT	Model	RCT	Model
Unemployment insurance	8.2 pp*** (1.8)	2.1 pp	3.0% (6.5)	3.3%
Cash transfer	4.1 pp** (1.8)	0.1 pp	-5.5% (7.8)	0.4%

- model correctly predicts larger response to UI than cash
- model understates migration response to UI and cash
- model predicts modest income gains, within the 68% CI of insignificant RCT effect

Information Provision: Barnett-Howell et al. (2026)

Sampling

RCT

- 17,000 rural households in 5 counties
- 30% of households within sampled villages treated

Model

- randomly treat 30% of rural households

Treatment

RCT

- info about urban labor market, 3 delivery methods
- measure beliefs about expected urban income
- information: beliefs ↑ 4%
- mentor: beliefs ↑ 10%
- group: beliefs ↑ 12%

Model

- information: $\gamma_{treated} = 1.04\gamma$
- mentor: $\gamma_{treated} = 1.10\gamma$
- group: experienced treated households set to $\gamma = 1$; inexperienced γ chosen to match average belief gain

Measurement

RCT

- endline: about 16 months after treatment
- migration and household income, treatment vs control

Model

- endline: 3 periods (18 months)
- urban residence and income changes relative to rural baseline income

Information Provision: Results

	Migration Rate Change		Income Change	
Treatment arm	RCT	Model	RCT	Model
Information	1.8 pp* (1.0)	0.9 pp	8.9%** (3.7)	1.2%
Mentor	2.4 pp** (1.2)	2.5 pp	9.3%** (4.2)	3.4%
Group	2.2 pp* (1.3)	4.3 pp	-2.8% (5.1)	4.9%

- information, mentor: model matches migration, understates income responses
- group: model overpredicts both migration and income
  - group more effective because of greater change in beliefs
  - heterogeneous engagement of experienced/inexperienced households not strong enough to suppress the income effect
- what could be missing?
  - economic spillovers via demand (e.g. slack)
  - greater heterogeneity in experience (not binary)

Quantitative Results

Long-Run Effects

- treatment effects grow beyond RCT endlines
- cash transfer *(Egger et al., 2022)* ▶ *dynamics*
  - migration is negligible at endline, but builds to 0.4 pp after 15 years
  - consumption effects decay slowly as the transfer is spent
- UI and cash transfer *(Miner, 2026)* ▶ *dynamics*
  - UI: about 40% of the 5-year migration effect is realized by endline
  - cash: migration builds for several years, then slowly decays
- information provision *(Barnett-Howell et al., 2026)* ▶ *dynamics*
  - 50–80% of long-run migration effects are realized by endline
  - gains continue growing for a few years as favorable shocks make migration attractive

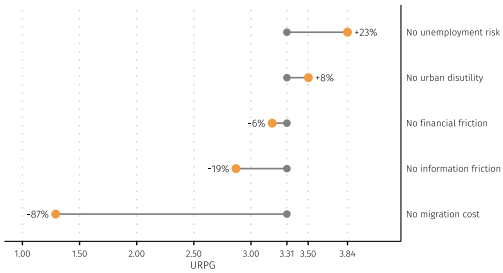
National Rollout

- apply info and UI treatments to all households

Scenario	Urban Share	URPG	Consumption
Baseline	30.5%	3.314	1.000
Barnett-Howell et al.: Individual	32.6%	3.242	1.018
Barnett-Howell et al.: Mentor	35.3%	3.178	1.040
Miner: Unemployment insurance	35.6%	3.197	1.064

- all interventions modestly boost urbanization, lower the urban–rural income gap
- mentor and UI generate the largest urbanization gains
  - but cost not equalized! UI much more expensive

Urban–Rural Income Gap Decomposition



Scenario	Urban Share	URPG	Consumption
Baseline	30.5%	3.314	1.000
No information friction	46.0%	2.869	1.078
No financial friction	36.2%	3.185	1.122
No unemployment risk	33.5%	3.844	1.212
No urban disutility	27.9%	3.502	0.973
No migration cost	77.3%	1.291	1.640

- information is the most important explicitly modeled friction for the income gap
- residual migration costs are the dominant barrier in the model

Conclusion

Summary and Next Steps

- **model**: model of rural–urban migration
  - info frictions, borrowing constraints, urban risk, migration costs
- **experiments**: reproduce several RCTs in one framework
  - matches Egger et al. cash transfer well
  - matches Miner UI and cash, Barnett-Howell et al. info less well
- **quantitative results**:
  - info frictions most important explicit friction behind URPG
  - most of the URPG is unexplained by explicit frictions, picked up by residual costs
- **further applications of the framework**:
  - compare cost-effectiveness of different interventions
  - slack mechanism for realistic local demand spillovers

Preferences and Experience

- CRRA utility over effective consumption:

$$u(c, l, x) = \frac{\tilde{c}^{1-\sigma} - 1}{1 - \sigma}, \qquad \tilde{c} = \frac{c}{1 + m_u \cdot \mathbf{1}(l = U, x = 0)}$$

- m_u: consumption-equivalent disutility for inexperienced urban migrants
  - inexperienced migrants in urban region discount consumption
- urban experience $x \in \{0, 1\}$:
  - gain experience ($x' = 1$) with prob. π_x per period in urban, stay experienced
  - lose experience ($x' = 0$) with prob. π_{-x} per period in rural, stay inexperienced
  - as in *Lagakos, Mobarak, Waugh (2023)*

back

Income, Productivity, and Employment

budget constraint: $c + a' + \mathbf{1}(l = R, l' = U) \cdot m_c = (1 + r)a + y(e, l, z_R, z_U)$

Employment

- urban: stochastic employment e
  - $e = 1 \to 0$: job separation rate λ_s
  - $e = 0 \to 1$: job finding rate λ_f
- rural: always employed ($e = 1$)

Income

$$y = \begin{cases} w_R z_R & \text{rural} \\ w_U z_U & \text{urban employed} \\ b & \text{urban unemployed} \end{cases}$$

rural beliefs: $\tilde{w}_U = \gamma w_U$, $\gamma \in (0, 1]$ ▶ *details*

Migration Cost

- pay m_c when moving $R \to U$

Bivariate AR(1) Productivity Process

$$\log z'_R = \rho_R \log z_R + \epsilon'_R$$
$$\log z'_U = \rho_U \log z_U + \epsilon'_U$$

$$(\epsilon'_R, \epsilon'_U) \sim N(\mathbf{0}, \mathbf{\Sigma})$$

- ρ_ϵ: cross-correlation of shocks

Assets

- save in one-period risk-free bond a'
  - return r
- borrowing constraint: $a' \geq \underline{a}$

▶ *production and GE*

back

Estimation

Assigned Parameters

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
$\underline{a}$	borrowing limit

Estimated Parameters

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$t = 6$ months

length of one model period

- maximizes available data moments for estimation

back

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$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$\beta = 0.98$

discount factor

$$V_{l \to l'} = \max_{a' \geq \underline{a}} u(c, R, x) + \beta \, \mathbb{E}\left[V_{l'}(a', z'_R, z'_U, x', e', \gamma)\right]$$

- standard for 6-month period length (from 0.96 for annual)

back

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$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$\sigma = 2$

coefficient of risk aversion in CRRA utility

$$u(c, l, x) = \frac{\tilde{c}^{1-\sigma} - 1}{1 - \sigma}$$

- standard

back

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ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$\alpha = 0.33$

capital share in production

$$Y_l = A_l K_l^{\alpha} L_l^{1-\alpha}$$

- labor share: 0.67
- standard

back

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ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$\delta = 0.051$

capital depreciation rate

$$r + \delta = \alpha A_l \left(K_l / L_l\right)^{\alpha - 1}$$

- standard for 6-month period length (from 0.10 for annual)

back

Estimation

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δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
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A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$A_R = 1$

rural total factor productivity

$$Y_R = A_R K_R^{\alpha} L_R^{1-\alpha}$$

- normalized

back

Estimation

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A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$\lambda_s = 0.06$

urban job separation rate

$$\Pr(e' = 0 \mid e = 1, l = U) = \lambda_s, \qquad \Pr(e' = 1 \mid e = 0, l = U) = \lambda_f$$

- **target moment**: 6-month unemployment hazard rate for urban migrants
- **data**: survey in *Barnett-Howell et al. (2026)*

back

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ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$\lambda_f = 0.42$

urban job finding rate

$$\Pr(e' = 0 \mid e = 1, l = U) = \lambda_s, \qquad \Pr(e' = 1 \mid e = 0, l = U) = \lambda_f$$

- **target moment**: 6-month employment hazard rate for urban migrants
- **data**: survey in *Barnett-Howell et al. (2026)*

back

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α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
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ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$m_u = 0.34$

consumption-equivalent utility cost of migration

$$\tilde{c} = \frac{c}{1 + m_u \cdot \mathbf{1}(l = U, x = 0)}$$

- **target moment**: share of income migrants are willing to give up to move back to rural, among those who are willing to give up some income to move back ("rural preference")
  - survey asked migrants what rural income would make them willing to move back
- **data**: survey in *Barnett-Howell et al. (2026)*

back

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$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$\pi_x = 0.45$

urban experience gaining rate

$$\Pr(x' = 1 \mid l = U, x = 0) = \pi_x, \qquad \Pr(x' = 0 \mid l = R, x = 1) = \pi_{-x}$$

- **target moment**: among rural households with rural preference, what share switches to urban preference within 6 months after migrating
  - location preference: require higher income to move to the other location
- **data**: survey in *Barnett-Howell et al. (2026)*

back

Estimation

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t	period length
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α	capital share
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λ_s	job separation
λ_f	job finding
m_u	migration disutility
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ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$\pi_{-x} = 0.68$

urban experience losing rate

$$\Pr(x' = 1 \mid l = U, x = 0) = \pi_x, \qquad \Pr(x' = 0 \mid l = R, x = 1) = \pi_{-x}$$

- **target moment**: among urban households with urban preference, what share switches to rural preference within 6 months after migrating
  - location preference: require higher income to move to the other location
- **data**: survey in *Barnett-Howell et al. (2026)*

back

Estimation

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ρ_U	urban persistence
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$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$\gamma = 0.56$

understatement of urban income by rural households

$$\tilde{w}_U = \gamma w_U$$

- **target moment**: avg expected Nairobi income by rural households as a share of actual Nairobi income, conditional on demographics
- **data**: *Barnett-Howell et al. (2026)*

back

Estimation

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$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$b = 0$

urban unemployment benefit

$$y(e, U, z_R, z_U) = \begin{cases} w_U z_U & e = 1 \\ b & e = 0 \end{cases}$$

- no formal unemployment insurance in Kenya at baseline

back

Estimation

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$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$\underline{a} = 0$
borrowing limit

$$c + a' + \mathbf{1}(l = R, l' = U) \cdot m_c = (1 + r)a + y(e, l, z_R, z_U), \qquad a' \geq \underline{a}$$

- no borrowing allowed at baseline

back

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$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$A_U = 2.02$

urban total factor productivity

$$Y_U = A_U K_U^{\alpha} L_U^{1-\alpha}$$

- **target moment**: exponentiated difference in mean log urban and rural incomes
- **data**: Kenya Integrated Household Budget Survey (KIHBS)

back

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$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$\rho_R = 0.68$

persistence of rural productivity

$$\log z'_R = \rho_R \log z_R + \epsilon'_R$$
$$\log z'_U = \rho_U \log z_U + \epsilon'_U$$
$$\text{SD}(\epsilon'_R) = \sigma_{\epsilon,R}, \quad \text{SD}(\epsilon'_U) = \sigma_{\epsilon,U}$$
$$\text{Corr}(\epsilon'_R, \epsilon'_U) = \rho_\epsilon$$

- **target moment**: standard deviation of log rural income
- **data**: Kenya Integrated Household Budget Survey (KIHBS)

back

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$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$\rho_U = 0.70$

persistence of urban productivity

$$\log z'_R = \rho_R \log z_R + \epsilon'_R$$
$$\log z'_U = \rho_U \log z_U + \epsilon'_U$$
$$\text{SD}(\epsilon'_R) = \sigma_{\epsilon,R}, \quad \text{SD}(\epsilon'_U) = \sigma_{\epsilon,U}$$
$$\text{Corr}(\epsilon'_R, \epsilon'_U) = \rho_\epsilon$$

- **target moment**: standard deviation of log urban income
- **data**: Kenya Integrated Household Budget Survey (KIHBS)

back

Estimation

Assigned Parameters

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
$\underline{a}$	borrowing limit

Estimated Parameters

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$\sigma_{\epsilon,R} = 1.12$

volatility of rural productivity shocks

$$\log z'_R = \rho_R \log z_R + \epsilon'_R$$
$$\log z'_U = \rho_U \log z_U + \epsilon'_U$$
$$\text{SD}(\epsilon'_R) = \sigma_{\epsilon,R}, \quad \text{SD}(\epsilon'_U) = \sigma_{\epsilon,U}$$
$$\text{Corr}(\epsilon'_R, \epsilon'_U) = \rho_\epsilon$$

- **target moment**: standard deviation of residual rural log income after controlling for persistence
- **data**: survey in *Barnett-Howell et al. (2026)*

back

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$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$\sigma_{\epsilon,U} = 1.04$

volatility of urban productivity shocks

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$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$\rho_\epsilon = 0.01$

cross-location correlation of productivity shocks within a household.

$$\log z'_R = \rho_R \log z_R + \epsilon'_R$$
$$\log z'_U = \rho_U \log z_U + \epsilon'_U$$
$$\text{SD}(\epsilon'_R) = \sigma_{\epsilon,R}, \quad \text{SD}(\epsilon'_U) = \sigma_{\epsilon,U}$$
$$\text{Corr}(\epsilon'_R, \epsilon'_U) = \rho_\epsilon$$

- **target moment**: slope from regressing urban log income on rural log income for similar households (clustered by observables)
- **model moment**: slope from regressing potential urban log income on potential rural log income within household
- **data**: *Barnett-Howell et al. (2026)*

back

Estimation

Assigned Parameters

t	period length
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α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
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ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$m_c = 3.05$

monetary migration cost, in units of mean annual household income

$$c + a' + \mathbf{1}(l = R, l' = U)\, m_c = (1 + r)a + y(e, l, z_R, z_U)$$

- **target moment**: share of Kenyan population living in cities
- **model moment**: share of households in urban region
- **data**: Kenya National Bureau of Statistics

back

Estimated Parameters and Target Moments

Parameter	Value	Target moment	Moment value	
			Data	Model
A_U	2.02	URPG	3.28	3.31
ρ_R	0.68	Standard deviation of log rural income	1.57	1.58
ρ_U	0.70	Standard deviation of log urban income	1.46	1.47
$\sigma_{\epsilon,R}$	1.12	Standard deviation of residual rural log income	1.14	1.15
$\sigma_{\epsilon,U}$	1.04	Standard deviation of residual urban log income	1.05	1.06
ρ_ϵ	0.01	Slope of urban vs rural log incomes	0.06	0.07
m_c	3.05[a]	Urban population share	0.32	0.30

[a] Here, m_c is converted into a multiple of mean annual labor income.

back

Cash Transfer: Egger et al. (2022)

Egger, Haushofer, Miguel, Niehaus, Walker (2022)

- **sample**: 10,500 poor households across 653 rural villages
  - randomly sampled from 653 villages within 1 county
- **intervention**: one-time huge unconditional cash transfer
  - 75% of mean annual household expenditure among recipients
  - 16% of annual village GDP
- **findings**:
  - expenditure ↑ 12%
  - no migration effect
  - large local spillovers: consumption ↑ similarly among treated and untreated
  - no change in local labor supply
  - local firms: sales ↑, wages ↑
- **requirements for model**:
  - financial frictions

back

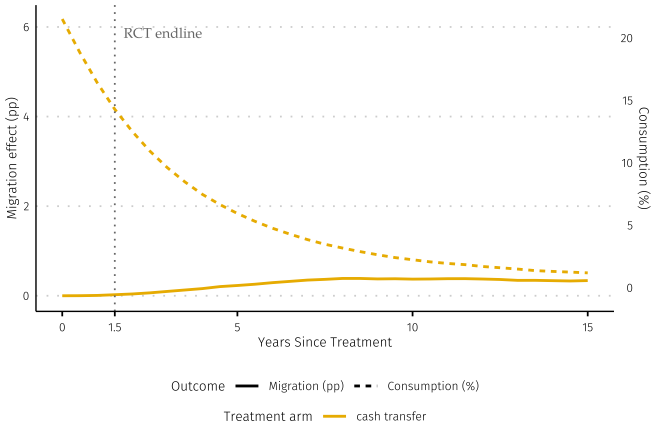
Cash Transfer: Egger et al. (2022), Alternative Experiments

Model treatment effects at endline (18 months)

Sampling	Migration Rate Change			Consumption Change		
	Low-asset, 1x	Representative, 1x	Low-asset, 2x	Low-asset, 1x	Representative, 1x	Low-asset, 2x
Cash transfer	0.0 pp	0.7 pp	0.1 pp	14.3%	5.5%	26.8%

back dynamics

Long-Run Effects: Cash Transfer, Egger et al. (2022)

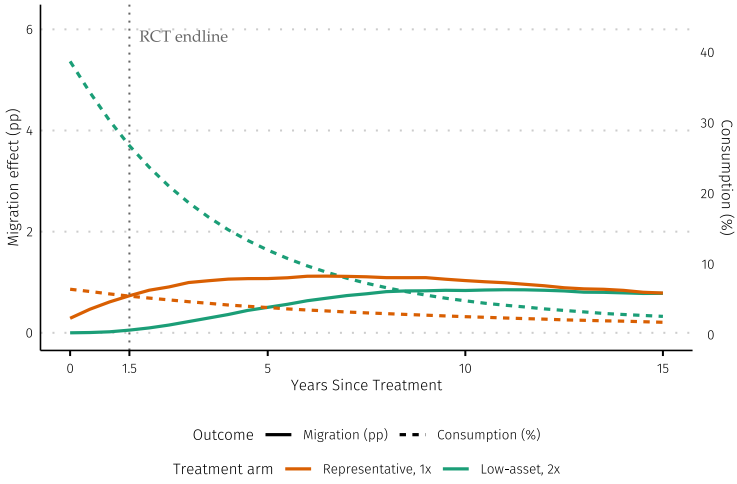


back

- consumption effects decay slowly as the transfer is spent
- migration is negligible at endline, but builds to 0.4 pp after 15 years
  - as more and more rural households experience shocks making migration attractive
- little migration response because targeted low-asset households: most can't afford the move even after the transfer

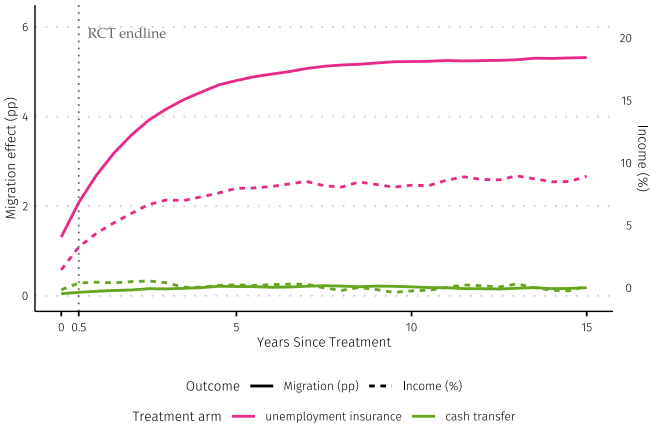
▶ *alternative experiments*

Long-Run Effects: Egger et al. (2022), Alternative Experiments



back endline effects

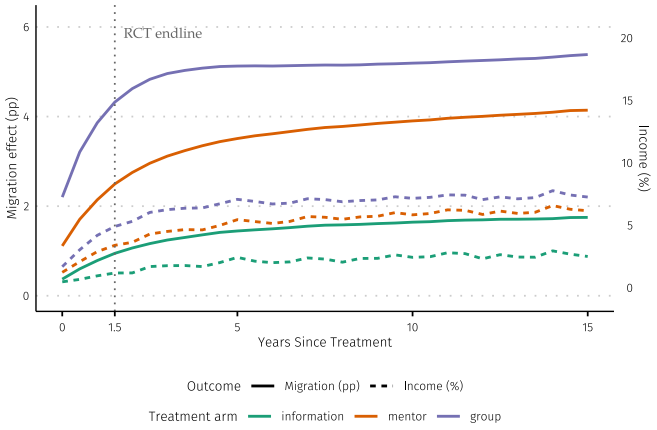
Long-Run Effects: UI and Cash Transfer, Miner (2026)



back

- UI: about 40% of the 5-year migration effect is realized by endline
- cash: migration builds for several years, then slowly decays

Long-Run Effects: Information Provision, Barnett-Howell et al. (2026)



back

- 50-80% of long-run migration effects are realized by endline
- gains continue growing for a few years as favorable shocks make migration attractive

Cash Transfer: Miner (2026)

Miner (2026)

- **sample**: about 1,300 young male workers
  - randomly sampled from villages within 1 county
- **intervention**: one-time unconditional cash transfer
- **findings**:
  - current Nairobi residence ↑ 4.1 p.p. (SE 1.8)
  - income: −5.5% (SE 7.8%), insignificant

back

Unemployment Insurance: Miner (2026)

Miner (2026)

- **sample**: about 1,300 young male workers
  - randomly sampled from villages within 1 county
- **intervention**: offer daily UI in Nairobi
  - $\sim$ daily urban wage
  - up to 15 days
- **findings**:
  - current Nairobi residence $\uparrow$ 8.2 p.p. (SE 1.8)
  - income: 3.0% (SE 6.5%), insignificant
- **requirements for model**:
  - unemployment risk

back

Information Provision: Barnett-Howell et al. (2026)

Barnett-Howell, Baseler, Ginn, Gordeev (2026)

- **sample**: 17,000 rural households
  - representatively sampled from 5 rural counties
- **intervention**: information about Nairobi income, labor market
- treatment arms:
  - **information**: individual delivery
  - **mentor**: individual + matched with urban mentor
  - **group**: delivery in village-wide meeting with group discussion

back

Information Provision: Barnett-Howell et al. (2026)

Barnett-Howell, Baseler, Ginn, Gordeev (2026)

- findings:
  - villagers underestimate conditional urban income by 44%
  - treatment → migration rate ↑ 2 p.p.
  - individual/mentor treatment → income ↑ approximately 9%
  - group treatment → no significant income change
  - group treatment primarily engages experienced migrants, who have low marginal returns (because of low migration cost)
  - large economic spillovers within village
- requirements for model:
  - information friction
  - heterogeneous migration cost (experience?) affecting marginal returns

back

Information Friction

- rural households perceive urban wage as $\tilde{w}_U = \gamma w_U$, where $\gamma \in (0, 1]$
  - $\gamma < 1$: rural households underestimate potential urban income
  - perceived income:

$$\tilde{y}(e, l, z_R, z_U, \gamma) = \begin{cases} w_R z_R & \text{if } l = R \\ \gamma w_U z_U & \text{if } l = U \text{ and } e = 1 \\ b & \text{if } l = U \text{ and } e = 0 \end{cases}$$

  - as in *Barnett-Howell et al. (2026)*
- rural→urban migrants are surprised by true urban wage
- urban households revert to misperception if they return to rural ▶ *belief dynamics*
  - otherwise everyone becomes well-informed in stationary equilibrium

back

Production

- production in each location l:

$$Y_l = A_l K_l^{\alpha} L_l^{1-\alpha}$$

- competitive input markets:

$$w_l = (1-\alpha) A_l (K_l/L_l)^{\alpha}$$

$$r + \delta = \alpha A_l (K_l/L_l)^{\alpha-1}$$

- perfect capital mobility
  - r equalized across regions

back

Equilibrium

1. household optimization given prices
   - rural households solve V_R with $\gamma < 1$
   - urban households solve V_U with $\gamma = 1$
2. firm optimization
3. market clearing:
   - labor: $L_l = \int z_l \, d\mu$
   - capital: $K_R + K_U = \int a \, d\mu$
4. stationary distribution

▶ *rural household problem* ▶ *urban household problem*

Recursive Formulation: Rural Household

rural household stays or migrates to urban:

$$V_R(a, z_R, z_U, x, \gamma) = \max\left\{V_{R\to R},\ V_{R\to U}\right\}$$

Stay in rural

$$V_{R\to R} = \max_{a'\geq \underline{a}} \quad u(c, R, x) + \beta\,\mathbb{E}\left[V_R(a', z_R', z_U', x', \gamma)\right]$$

$$c + a' = (1+r)a + \tilde{y}(1, R, z_R, z_U, \gamma)$$

Migrate to urban

$$V_{R\to U} = \max_{a'\geq \underline{a}} \quad u(c, R, x) + \beta\,\mathbb{E}\left[V_U(a', z_R', z_U', x', e', \gamma)\right]$$

$$c + a' + m_c = (1+r)a + \tilde{y}(1, R, z_R, z_U, \gamma)$$

back

Recursive Formulation: Urban Household

urban household stays or migrates to rural:

$$V_U(a, z_R, z_U, x, e, \gamma) = \max\left\{V_{U\to U},\ V_{U\to R}\right\}$$

Stay in urban

$$V_{U\to U} = \max_{a'\geq \underline{a}} \quad u(c, U, x) + \beta\,\mathbb{E}\left[V_U(a', z_R', z_U', x', e', \gamma)\right]$$

$$c + a' = (1+r)a + \tilde{y}(e, U, z_R, z_U, \gamma)$$

Migrate to rural

$$V_{U\to R} = \max_{a'\geq \underline{a}} \quad u(c, U, x) + \beta\,\mathbb{E}\left[V_R(a', z_R', z_U', x', \gamma)\right]$$

$$c + a' + m_c = (1+r)a + \tilde{y}(e, U, z_R, z_U, \gamma)$$

back

Belief Dynamics of Urban Households

1. **Income beliefs:** while in U, households use V with $\gamma = 1$
   - do not anticipate forgetting the true urban wage after returning to R
2. **Migration cost beliefs:** while in U, households use V with $m_c = 0$
   - do not anticipate facing migration cost after returning to R for repeat $R \to U$ moves
   - allows the stationary equilibrium to hit URPG, urban share moments

back